University of Birmingham Dubai

Fluid Mechanics and Energy Transfer

Name of Experiment: FMET2 - Momentum Principle and Impact of Jets

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Date & Time of Lab: Not sure

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Executive Summary

This report discusses the FMET2 lab experiment, which tested the momentum principle using a water jet hitting two different surfaces: a flat plate and a hemispherical target. The main goal was to measure the force made by the jet and compare it to the force predicted by the theory. A lever system was used to measure how far the target moved when hit by the water. Water was collected to find the flow rate, and from there, we calculated the jet speed. These values were used to work out the experimental force and theoretical force.

For the flat plate, where the water jet deflects at 90 degrees, the experimental results closely matched the theoretical predictions. Graph 1 showed that the measured force almost followed the ideal y = x line, meaning minimal energy loss. However, for the hemispherical target, where the jet deflects at 180 degrees, the experimental forces were less than the theoretical values. This is mainly because the curved surface caused more energy loss due to turbulence and splashing. Graph 2 showed greater differences between experimental and theoretical values, proving that real-world effects impacted the results.

In conclusion, the FMET2 lab clearly showed how momentum transfer happens between a moving jet and different shapes. The flat plate gave results closer to theory, while the hemisphere showed bigger energy losses. The experiment proved the momentum principle works well but also showed that in real applications, extra factors like turbulence must be considered.

1. Theory

When a water jet hits a surface, it changes direction, and its momentum changes. The force on the surface depends on how much momentum the jet loses. For a flat plate, force is $F = \rho A u^2$. For a hemisphere, force is $F = 2\rho A u^2$. These formulas assume steady, lossless flow of an incompressible fluid.

2. Experimental Method

A water jet was aimed vertically at two different targets attached to a lever. The lever's movement was measured to find force. Flow rate was found by collecting water over time. Velocity was calculated from the flow rate and nozzle size. Forces were compared with theoretical values from the momentum principle.

3. Tables

3.1 Flat Plate Data

x (m)	Q (L/s)	Experimental Force (N)	Jet Velocity (m/s)	Theoretical Force (N)
0.081	0.510	3.18	6.44	3.26
0.076	0.495	2.98	6.25	3.07
0.068	0.474	2.67	5.98	2.81
0.057	0.442	2.24	5.57	2.43
0.047	0.401	1.84	5.04	1.99

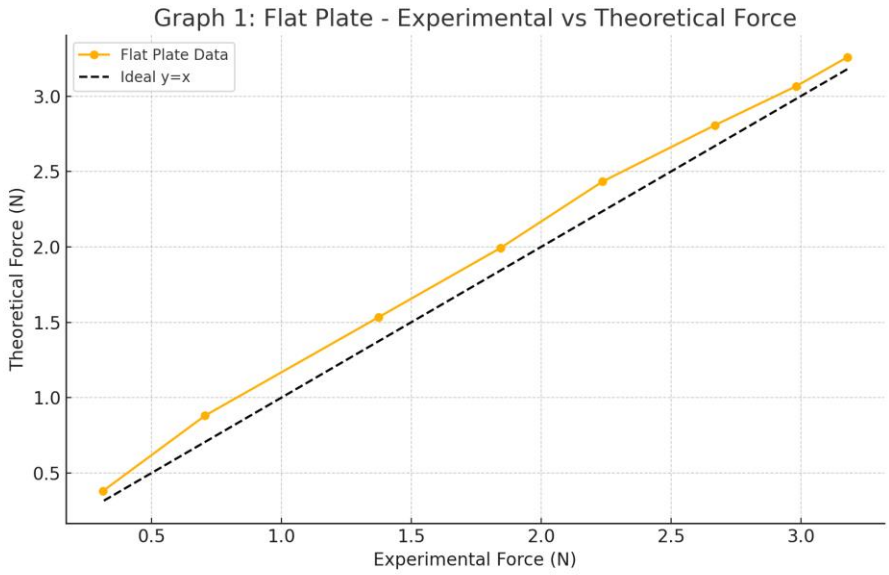
0.035	0.353	1.37	4.42	1.53
0.018	0.271	0.71	3.35	0.88
0.008	0.185	0.31	2.20	0.38

3.2 Hemispherical Target Data

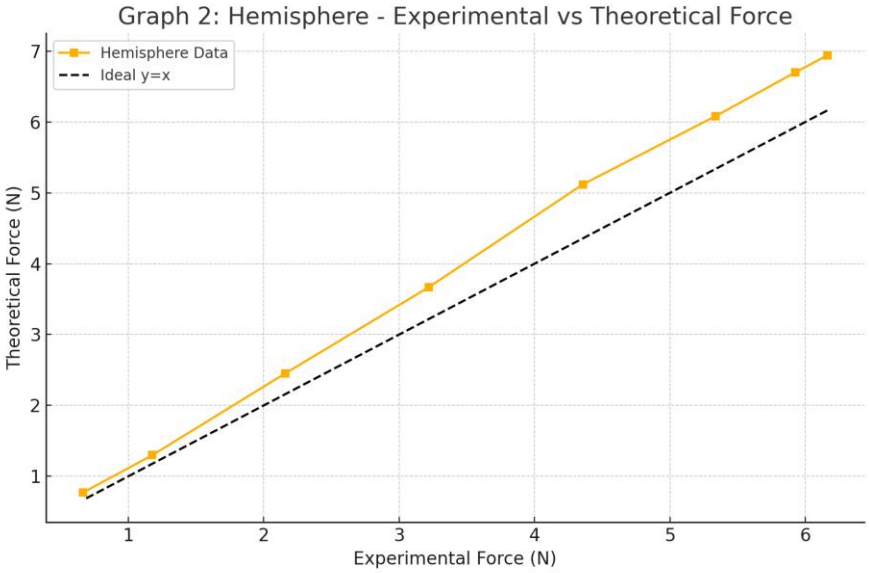
x (m)	Q (L/s)	Experimental Force (N)	Jet Velocity (m/s)	Theoretical Force (N)
0.157	0.526	6.16	6.65	6.94
0.151	0.517	5.93	6.53	6.70
0.136	0.493	5.34	6.22	6.08
0.111	0.453	4.36	5.71	5.12
0.082	0.385	3.22	4.83	3.67
0.055	0.317	2.16	3.95	2.45
0.030	0.235	1.18	2.88	1.30
0.017	0.186	0.67	2.22	0.77

4. Graphs

Graph 1: Flat Plate - Experimental vs Theoretical Force



Graph 2: Hemisphere - Experimental vs Theoretical Force



5. Results and Discussion

For the flat plate target, the experimental forces closely matched the theoretical forces predicted by the momentum principle. Graph 1 showed a near-linear relationship along the ideal y = x line, confirming that the jet's momentum transfer was efficient and energy losses were minimal. Minor deviations observed can be attributed to small errors in force measurements, fluctuations in water flow, and slight inaccuracies in flow rate determination. The results validate that, for flat surfaces, the theoretical assumption of steady and direct momentum change holds true.

For the hemispherical target, the experimental forces were consistently lower than the theoretical predictions. Graph 2 showed a clear deviation from the y = x line, especially at higher flow rates. The hemisphere's curved surface caused the water jet to spread, leading to splash-back, turbulence, and additional energy losses. These real-world effects reduced the effective momentum transfer to the surface, meaning the actual force recorded was smaller than ideal. Measurement uncertainties, air resistance, and surface roughness could also have contributed to the deviations.

Overall, the FMET2 experiment successfully demonstrated the momentum principle and the practical impact of different surface shapes on jet force. While the flat plate showed almost ideal behavior, the hemisphere highlighted how real-world factors like turbulence can significantly affect force transfer. The findings stress the importance of considering such factors in real engineering applications involving fluid-structure interactions, such as turbine blade design or nozzle impacts.